CS182: Introduction to Machine Learning
Lecture 02: Linear Algebra

Lecture Quiz and Answer Copy

Scope: Linear_Algebra.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Name: _____ **Student ID:** _____

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Answer sheet — Section A

1	2	3	4	5	6

7	8	9	10	11	12

Answer sheet — Section B

13	14	15	16	17	18

19	20	21	22	23	24

Section A. True/False 12 marks

1. Under the lecture convention, the gradient of a scalar function of $x \in \mathbb{R}^n$ is an $n \times 1$ column vector.

2. If f is twice continuously differentiable on an open neighborhood of x, then its Hessian at x is symmetric.

3. Every pair of eigenvectors of a real symmetric matrix is automatically orthogonal, including pairs with the same eigenvalue.

4. A square matrix has an inverse if and only if its determinant is nonzero.

5. The correct transpose rule for a product is $(AB)^T = A^T B^T$.

6. Every real symmetric positive semidefinite matrix is invertible.

7. A real $m \times n$ matrix represents a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$.

8. For $x, y \in \mathbb{R}^3$, the outer product xy^T is a scalar.

9. Every invertible real symmetric matrix has only positive eigenvalues.

10. If two real vectors have zero inner product, at least one of them must be the zero vector.

11. If $A \in \mathbb{R}^{2\times 4}$ has full rank, then $Ax = 0$ has only the zero solution.

12. If both AB and BA are defined and have the same shape, they must be equal.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

13. Let A, B be invertible real square matrices of the same size. Select every valid formula.

(A) $\det(AB) = \det(A) + \det(B)$.

(B) $(A^T)^{-1} = (A^{-1})^T$.

(C) $(AB)^{-1} = A^{-1}B^{-1}$ for all such matrices.

(D) $(2A)^{-1} = 2A^{-1}$.

14. For $x = (-2, 1, 2)^T$, select every correct norm value.

(A) $\|x\|_\infty = 3$.

(B) $\|x\|_2 = 3$.

(C) $\|x\|_1 = 5$.

(D) $\|x\|_1 = |-2+1+2| = 1$.

15. A Cholesky factorization is given as $C = T^T T$, where T is a nonsingular upper triangular real matrix. Select every correct conclusion.

(A) C is positive definite.

(B) The same factor always also satisfies $C = TT^T$.

(C) The diagonal entry c_{ii} always equals the squared length of row i of T.

(D) $\det(C) = \prod_i t_{ii}$ always holds.

16. Consider a proposed eigendecomposition $A = Q\Lambda Q^{-1}$, with diagonal Λ. Select every correct statement.

(A) The columns of Q must be linearly independent.

(B) Normalizing every column of Q to length one is sufficient to replace the inverse by its transpose.

(C) Every real square matrix has such a decomposition with a real invertible Q and a real diagonal Lambda.

(D) A real symmetric matrix permits an orthogonal choice of Q and real diagonal entries in the decomposition.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

17. For fixed $A \in \mathbb{R}^{m\times n}$, let $f(u, v) = u^T Av$, with $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$. Select every correct statement.

(A) When v is fixed, $\nabla_u f = Av$.

(B) When u is fixed, $\nabla_v f = A^T u$.

(C) When v is fixed, the Hessian with respect to u is the zero $m \times m$ matrix.

(D) When u is fixed, $\nabla_v f = Au$ regardless of dimensions.

18. Let $Q \in \mathbb{R}^{n\times n}$ satisfy $Q^T Q = I$. Select every statement that holds for all real $x, y \in \mathbb{R}^n$.

(A) If x and y are nonzero, their angle is preserved by Q.

(B) $\|Qx\|_1 = \|x\|_1$.

(C) $(Qx)^T(Qy) = x^T y$.

(D) $\|Qx - Qy\|_2 = \|x - y\|_2$.

19. Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$. Select every correct trace identity.

(A) $\operatorname{tr}(A^T A) = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2$.

(B) For square C, $\operatorname{tr}(C^T) = \operatorname{tr}(C)$.

(C) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

(D) When A, B are square, $\operatorname{tr}(AB) = \operatorname{tr}(A)\operatorname{tr}(B)$.

20. Let C be a real symmetric positive definite matrix and $g(C) = \log\det C$. Matrix gradients use $dg = \operatorname{tr}((\nabla_C g)^T\, dC)$. Select every correct statement.

(A) The real logarithm of the determinant is defined for every invertible real matrix.

(B) For a symmetric perturbation H, the directional derivative is $\operatorname{tr}(C^{-1}H)$.

(C) For positive scalar c, if $C = cI_n$, then $g(C) = n \log c$.

(D) $\nabla_C g = C^{-1}$ at such a matrix.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

21. Let $A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \end{pmatrix}$, with columns a_1, a_2, a_3, and $x = (2, -1, 1)^T$. Select every correct statement.

(A) $Ax = 2a_1 - a_2 + a_3$.

(B) Every vector Ax, as x varies, lies in the span of the columns of A.

(C) $Ax = (0, 1)^T$.

(D) $A^T x$ is defined for this x.

22. Let $f(x_1, x_2) = x_1^2 x_2 + 3x_2^2$. Select every correct statement.

(A) $\nabla f(1, 1) = (2, 7)^T$.

(B) $\nabla^2 f(1, 0) = \begin{pmatrix} 0 & 2 \\ 2 & 6 \end{pmatrix}$ is indefinite.

(C) The Hessian is constant because f is a polynomial.

(D) $\nabla^2 f(1, 1) = \begin{pmatrix} 2 & 2 \\ 2 & 6 \end{pmatrix}$ is positive definite.

23. Let A, B be real symmetric PSD matrices of the same size. Select every statement that always holds.

(A) $x^T ABx \geq 0$ for every real vector x.

(B) A+B is PSD.

(C) AB is symmetric.

(D) For every real matrix R with compatible dimensions, $R^T AR$ is PSD.

24. Let $x, y \in \mathbb{R}^n$ both be nonzero. Select every correct statement about outer products.

(A) $xy^T = yx^T$ for all such x, y.

(B) xy^T has rank one.

(C) xx^T is symmetric positive semidefinite.

(D) $\mathrm{tr}(xy^T) = x^T y$.

Section C. Proofs and Formula Derivations

25. The nullspace and definiteness of a Gram matrix **[10 marks]**

(a) For any real $A \in \mathbb{R}^{m \times n}$, prove $\text{null}(A^T A) = \text{null}(A)$. Your proof must establish both inclusions without assuming A is square. **[3 marks]**

(b) Use the result to prove that $A^T A$ is PSD, that its rank equals the rank of A, and that it is PD if and only if A has full column rank. Explain the case $m < n$. **[4 marks]**

(c) For $\mu > 0$, prove $A^T A + \mu I_n$ is PD for every A. Give a lower bound on its quadratic form and explain why its inverse exists. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

26. Differentiate a nonsymmetric quadratic and a bilinear form **[10 marks]**

(a) Let $A \in \mathbb{R}^{n\times n}$ be fixed, with no symmetry assumption, and $f(x) = x^T Ax$. Expand $f(x+h) - f(x)$ and derive its gradient and Hessian. Explain precisely why the formula $2Ax$ need not be correct.**[4 marks]**

(b) Decompose $A = S + K$, where $S = (A + A^T)/2$ and $K = (A - A^T)/2$. Prove $x^T Kx = 0$, and show that knowledge of $x^T Ax$ for every real x cannot identify K. **[3 marks]**

(c) For fixed $B \in \mathbb{R}^{m\times n}$, derive the first differential of $g(u, v) = u^T Bv$ and both partial gradients. Then state the derivative along differentiable paths $u(t), v(t)$. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

27. Orthogonal eigenspaces and quadratic-form bounds **[10 marks]**

(a) Let $A = A^T$, $Au = \alpha u$, and $Av = \beta v$, with nonzero eigenvectors and $\alpha \neq \beta$. Prove $u^T v = 0$. Explain what must be changed when the eigenvalues are equal. **[3 marks]**

(b) Using the spectral theorem for real symmetric matrices, prove that A is PSD if and only if all its eigenvalues are nonnegative, and PD if and only if all are positive. Also establish the relation between rank and nonzero eigenvalues. **[4 marks]**

(c) For nonzero x, derive the Rayleigh quotient bounds $\lambda_{\min} \leq x^T Ax/(x^T x) \leq \lambda_{\max}$. Characterize equality at each endpoint. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

28. Build an orthogonal projector from an orthonormal basis **[10 marks]**

(a) Let $Q \in \mathbb{R}^{m \times r}$ have orthonormal columns, with $1 \leq r \leq m$. Define $P = QQ^T$. Prove $P^T = P$, $P^2 = P$, and $Pz = z$ for every $z \in \text{col}(Q)$. **[3 marks]**

(b) For an arbitrary $y \in \mathbb{R}^m$, show that $y = Py + (I - P)y$ is a decomposition into a vector in $\text{col}(Q)$ and a vector orthogonal to that space. Derive a Pythagorean norm identity. **[4 marks]**

(c) Prove that Py is the unique nearest vector to y in $\text{col}(Q)$. Then prove $\|Py\|_2 \leq \|y\|_2$ and give its equality condition. **[3 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

29. A rectangular sensor map: rank, Cholesky, and residual geometry [12 marks]

(a) A sensor model is Az, where $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$, and the observed vector is $b = (2, 0, 1)^T$. Compute $G = A^T A$ and $A^T b$, establish the column rank of A, and decide whether G is PD. **[3 marks]**

(b) Find an upper triangular T with positive diagonal such that $G = T^T T$, and verify the product. **[3 marks]**

(c) Find z_* by solving $Gz_* = A^T b$. Compute $r = b - Az_*$, verify $A^T r = 0$, and use a squared-distance expansion to show that this solution gives the unique best least-squares fit. **[4 marks]**

(d) Find the eigenvalues of G and an orthonormal eigenbasis. Express $\|Ad\|_2^2$ using the coordinates of d in that basis. **[2 marks]**

Student response

(a)

(b)

(c)

(d)

Section D. Applications

30. A nonsymmetric representation of a quadratic objective **[12 marks]**

(a) Let $f(x) = \frac{1}{2}x^T Bx - c^T x$, where $B = \begin{pmatrix} 4 & 5 \\ -1 & 6 \end{pmatrix}$, $c = (6, 8)^T$, and $x \in \mathbb{R}^2$. Compute its gradient and Hessian. Identify the symmetric matrix governing its quadratic part. **[3 marks]**

(b) Find the eigenvalues of the Hessian and decide whether it is PD, PSD but not PD, or indefinite. Give a quadratic-form verification of strict positivity as an independent check. **[3 marks]**

(c) Solve for the stationary point, compute its objective value, and prove that it is the unique global minimizer. **[3 marks]**

(d) Start at $x_0 = 0$ and take one gradient step $x_1 = x_0 - \eta \nabla f(x_0)$, with $\eta > 0$. Find the complete range of step sizes that strictly decreases the objective on this first step. Also find the best step size along this direction. **[3 marks]**

Student response

(a)

(b)

(c)

(d)